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JEE MAIN PHYSICS DPP: DIMENSIONAL ANALYSIS

Topic: Dimensions of Physical Quantities & Dimensional Formulae | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** The dimensional formula of Planck's constant (h) is identical to that of:
- Linear momentum
 - Angular momentum
 - Rotational kinetic energy
 - Work
- Q2.** If ε_0 denotes the permittivity of free space and E is the electric field, the dimensional formula of $\frac{1}{2}\varepsilon_0 E^2$ (energy density) is:
- $[M^1 L^{-1} T^{-2}]$
 - $[M^1 L^2 T^{-2}]$
 - $[M^1 L^{-2} T^{-1}]$
 - $[M^1 L^1 T^{-2}]$
- Q3.** Which of the following pairs of physical quantities does **not** have identical dimensional formulae?
- Torque and Work
 - Stress and Young's modulus
 - Electric flux and Electric field
 - Impulse and Linear momentum
- Q4.** In the formula $X = 3YZ^2$, X and Z have dimensions of capacitance and magnetic induction respectively. The dimensional formula of Y in SI units is:
- $[M^{-3} L^{-2} T^8 A^4]$
 - $[M^{-1} L^{-2} T^4 A^2]$
 - $[M^{-3} L^{-2} T^4 A^1]$
 - $[M^{-2} L^{-3} T^8 A^4]$
- Q5.** The dimensional formula for the coefficient of dynamic viscosity (η) is:
- $[M^1 L^{-1} T^{-1}]$
 - $[M^1 L^1 T^{-1}]$
 - $[M^1 L^{-2} T^{-2}]$
 - $[M^1 L^{-1} T^{-2}]$
- Q6.** The magnetic permeability of free space (μ_0) has the dimensional formula:
- $[M^1 L^1 T^{-2} A^{-2}]$
 - $[M^1 L^2 T^{-2} A^{-1}]$
 - $[M^1 L^1 T^{-1} A^{-2}]$
 - $[M^1 L^2 T^{-1} A^{-2}]$
- Q7.** If L, C , and R represent inductance, capacitance, and resistance respectively, which of the following combinations has dimensions of frequency?
- $\frac{R}{L}$
 - $\frac{1}{RC}$
 - $\frac{1}{\sqrt{LC}}$
 - All of the above
- Q8.** The dimensional formula of self-inductance (L) is:
- $[M^1 L^2 T^{-2} A^{-2}]$
 - $[M^1 L^1 T^{-2} A^{-2}]$
 - $[M^1 L^2 T^{-1} A^{-2}]$
 - $[M^1 L^2 T^{-2} A^{-1}]$
- Q9.** The dimensional formula of thermal conductivity (K) is:
- $[M^1 L^1 T^{-3} K^{-1}]$
 - $[M^1 L^2 T^{-3} K^{-1}]$
 - $[M^1 L^1 T^{-2} K^{-1}]$
 - $[M^1 L^{-1} T^{-3} K^{-1}]$
- Q10.** In the gas equation $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, the dimensional formula of the constant a is:
- $[M^1 L^5 T^{-2}]$
 - $[M^1 L^3 T^{-2}]$
 - $[M^1 L^{-1} T^{-2}]$
 - $[M^1 L^2 T^{-2}]$
- Q11.** Which of the following groups of physical quantities consists exclusively of dimensionless parameters?
- Strain, Poisson's ratio, Relative permeability
 - Refractive index, Dielectric constant, Specific heat
 - Angle, Solid angle, Gravitational potential
 - Reynolds number, Mach number, Surface tension
- Q12.** The dimensional formula of electric potential (V) is:
- $[M^1 L^2 T^{-3} A^{-1}]$
 - $[M^1 L^1 T^{-3} A^{-1}]$
 - $[M^1 L^2 T^{-2} A^{-1}]$
 - $[M^1 L^2 T^{-3} A^{-2}]$
- Q13.** The dimensional formula of the universal gravitational constant (G) is:
- $[M^{-1} L^3 T^{-2}]$
 - $[M^1 L^2 T^{-2}]$
 - $[M^{-1} L^2 T^{-1}]$
 - $[M^1 L^3 T^{-2}]$

- Q14.** If force F , length L , and time T are taken as fundamental physical quantities, the dimensional formula of mass is:
- $[F^1 L^{-1} T^2]$
 - $[F^1 L^1 T^{-2}]$
 - $[F^1 L^{-1} T^{-1}]$
 - $[F^1 L^2 T^{-2}]$
- Q15.** The dimensional formula for the Stefan-Boltzmann constant (σ) is:
- $[M^1 L^0 T^{-3} K^{-4}]$
 - $[M^1 L^1 T^{-3} K^{-4}]$
 - $[M^1 L^2 T^{-2} K^{-4}]$
 - $[M^1 L^0 T^{-2} K^{-4}]$
- Q16.** The quantity $\frac{e^2}{4\pi\epsilon_0 hc}$, where e is electronic charge, h is Planck's constant, and c is speed of light, has the dimension of:
- $[M^0 L^0 T^0]$ (Dimensionless)
 - $[M^1 L^1 T^{-1}]$
 - $[M^1 L^2 T^{-2}]$
 - $[M^0 L^1 T^{-1}]$
- Q17.** The dimensional formula of solar constant (energy received per unit area per second) is:
- $[M^1 L^0 T^{-3}]$
 - $[M^1 L^2 T^{-2}]$
 - $[M^1 L^1 T^{-2}]$
 - $[M^1 L^0 T^{-2}]$
- Q18.** The magnetic dipole moment (M) of a current loop has the dimensional formula:
- $[M^0 L^2 T^0 A^1]$
 - $[M^0 L^1 T^0 A^1]$
 - $[M^1 L^2 T^{-1} A^1]$
 - $[M^0 L^2 T^{-1} A^1]$
- Q19.** In the relation $v = at + \frac{b}{t+c}$, where v is velocity and t is time, the dimensional formulae of a , b , and c are respectively:
- $[LT^{-2}]$, $[L]$, $[T]$
 - $[LT^{-1}]$, $[L]$, $[T]$
 - $[LT^{-2}]$, $[LT^{-1}]$, $[T]$
 - $[L^2 T^{-2}]$, $[L]$, $[T^2]$
- Q20.** The dimensional formula of the universal gas constant (R) is:
- $[M^1 L^2 T^{-2} K^{-1} \text{mol}^{-1}]$
 - $[M^1 L^1 T^{-2} K^{-1} \text{mol}^{-1}]$
 - $[M^1 L^2 T^{-3} K^{-1} \text{mol}^{-1}]$
 - $[M^1 L^2 T^{-2} K^1 \text{mol}^{-1}]$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(a)	11	(a)	16	(a)
2	(a)	7	(d)	12	(a)	17	(a)
3	(c)	8	(a)	13	(a)	18	(a)
4	(a)	9	(a)	14	(a)	19	(a)
5	(a)	10	(a)	15	(a)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

$$E = h\nu \implies [h] = \frac{[E]}{[\nu]} = \frac{[M^1 L^2 T^{-2}]}{[T^{-1}]} = [M^1 L^2 T^{-1}].$$

Angular momentum $L = mvr$ has dimensions:

$$[L] = [M][LT^{-1}][L] = [M^1 L^2 T^{-1}]$$

Thus, $[h] = [L]$.

Sol 2. Correct Option: (a)

$\frac{1}{2}\epsilon_0 E^2$ is electrostatic energy per unit volume:

$$\left[\frac{\text{Energy}}{\text{Volume}}\right] = \frac{[M^1 L^2 T^{-2}]}{[L^3]} = [M^1 L^{-1} T^{-2}]$$

Sol 3. Correct Option: (c)

Electric flux $\Phi_E = E \cdot A \implies [\Phi_E] = [E][L^2] \neq [E]$.

All other given pairs have matching dimensions:

- Torque & Work: $[M^1 L^2 T^{-2}]$
- Stress & Modulus: $[M^1 L^{-1} T^{-2}]$
- Impulse & Momentum: $[M^1 L^1 T^{-1}]$.

Sol 4. Correct Option: (a)

Capacitance $[X] = [M^{-1} L^{-2} T^4 A^2]$.

Magnetic induction $[Z] = [M^1 L^0 T^{-2} A^{-1}]$.

From $X = 3YZ^2$:

$$[Y] = \frac{[X]}{[Z]^2} = \frac{[M^{-1} L^{-2} T^4 A^2]}{[M^2 T^{-4} A^{-2}]} = [M^{-3} L^{-2} T^8 A^4]$$

Sol 5. Correct Option: (a)

From Stokes' law $F = 6\pi\eta rv$:

$$[\eta] = \frac{[F]}{[r][v]} = \frac{[M^1 L^1 T^{-2}]}{[L][LT^{-1}]} = [M^1 L^{-1} T^{-1}]$$

Sol 6. Correct Option: (a)

From Biot-Savart Law, $B = \frac{\mu_0 I}{2\pi r} \implies \mu_0 = \frac{2\pi r B}{I}$.

Since $[B] = [M^1 T^{-2} A^{-1}]$:

$$[\mu_0] = \frac{[L][M^1 T^{-2} A^{-1}]}{[A]} = [M^1 L^1 T^{-2} A^{-2}]$$

Sol 7. Correct Option: (d)

$$\bullet \left[\frac{R}{L}\right] = \frac{1}{[\text{Time constant}]} = [T^{-1}]$$

$$\bullet \left[\frac{1}{RC}\right] = \frac{1}{[\tau]} = [T^{-1}]$$

$$\bullet \left[\frac{1}{\sqrt{LC}}\right] = [\omega_0] = [T^{-1}].$$

All three express frequency $[T^{-1}]$.

Sol 8. Correct Option: (a)

$$\text{Induced EMF } |\mathcal{E}| = L \frac{dI}{dt} \implies [L] = \frac{[\mathcal{E}][dt]}{[dI]}.$$

$$[\mathcal{E}] = [M^1 L^2 T^{-3} A^{-1}].$$

$$[L] = \frac{[M^1 L^2 T^{-3} A^{-1}][T]}{[A]} = [M^1 L^2 T^{-2} A^{-2}]$$

Sol 9. Correct Option: (a)

$$\text{Heat flow } \frac{dQ}{dt} = \frac{KA\Delta T}{x} \implies K = \frac{(dQ/dt)x}{A\Delta T}.$$

$$[K] = \frac{[M^1 L^2 T^{-3}][L]}{[L^2][K]} = [M^1 L^1 T^{-3} K^{-1}]$$

Sol 10. Correct Option: (a)

By principle of dimensional homogeneity, terms added must have identical dimensions:

$$[P] = \left[\frac{a}{V^2}\right] \implies [a] = [P][V]^2$$

$$[a] = [M^1 L^{-1} T^{-2}][L^3]^2 = [M^1 L^5 T^{-2}]$$

Sol 11. Correct Option: (a)

Strain ($\Delta L/L$), Poisson's ratio ($\Delta d/d \div \Delta L/L$), and Relative Permeability ($\mu_r = \mu/\mu_0$) are pure dimensionless ratios ($[M^0 L^0 T^0]$).

Sol 12. Correct Option: (a)

Electric potential $V = \frac{W}{q}$:

$$[V] = \frac{[M^1 L^2 T^{-2}]}{[A^1 T^1]} = [M^1 L^2 T^{-3} A^{-1}]$$

Sol 13. Correct Option: (a)

$$F = \frac{Gm_1 m_2}{r^2} \implies G = \frac{Fr^2}{m^2}.$$

$$[G] = \frac{[M^1 L^1 T^{-2}][L^2]}{[M^2]} = [M^{-1} L^3 T^{-2}]$$

Sol 14. Correct Option: (a)

$$\text{Force } [F] = [M^1 L^1 T^{-2}] \implies [M] = [F^1 L^{-1} T^2].$$

Sol 15. Correct Option: (a)

$$\text{Stefan-Boltzmann law } E = \sigma T^4 \implies \sigma = \frac{E}{T^4},$$

where E is power per unit area:

$$[E] = \frac{[M^1 L^2 T^{-3}]}{[L^2]} = [M^1 T^{-3}]$$

$$[\sigma] = \frac{[M^1 T^{-3}]}{[K^4]} = [M^1 L^0 T^{-3} K^{-4}]$$

Sol 16. Correct Option: (a)

$$\text{This is the fine structure constant } \alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137}, \text{ which is a pure dimensionless constant } ([M^0 L^0 T^0]).$$

Sol 17. Correct Option: (a)

$$\text{Solar constant } S = \frac{\text{Energy}}{\text{Area} \times \text{Time}}:$$

$$[S] = \frac{[M^1 L^2 T^{-2}]}{[L^2][T]} = [M^1 L^0 T^{-3}]$$

Sol 18. Correct Option: (a)

$$\text{Magnetic dipole moment } M = I \times A:$$

$$[M] = [A][L^2] = [M^0 L^2 T^0 A^1]$$

Sol 19. Correct Option: (a)

$$\text{By principle of homogeneity: } \bullet [at] = [v] \implies$$

$$[a] = \frac{[LT^{-1}]}{[T]} = [LT^{-2}]$$

$$\bullet [t + c] \implies [c] = [T]$$

$$\bullet \left[\frac{b}{t + c} \right] = [v] \implies [b] = [v][T] = [LT^{-1}][T] = [L].$$

Sol 20. Correct Option: (a)

$$\text{From ideal gas equation } PV = nRT:$$

$$[R] = \frac{[P][V]}{[n][T]} = \frac{[M^1 L^2 T^{-2}]}{[\text{mol}][K]}$$

$$= [M^1 L^2 T^{-2} K^{-1} \text{mol}^{-1}]$$

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